ABDULLAH AL BASHIT, Ph.D.

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Website | LinkedIn | GitHub | Google Scholar

(U.S. Permanent Resident, authorized to work in the U.S.)

PROFESSIONAL EXPERIENCE

Computational Imaging Collaborator *(Neurovascular Segmentation)* Apr 2026 – Present

Northeastern University, Dept. of Bioengineering, Boston, MA

- Created an end-to-end training and inference pipeline to segment blood-vessel structures in noisy 3D fluorescence microscopy images, handling weak contrast, uneven focus, broken structures, and limited annotated data.
- Fine-tuned an Attention U-Net with a pretrained ResNet34 encoder, using attention-gated skip connections to infer latent vessel structure, guided by engineered image features and augmentation.
- Trained and validated the model on an HPC cluster with 5-fold cross-validation, checkpointing, W&B tracking, attention-map review, and TP/FP/FN failure analysis, achieving approximately 91% Dice on held-out test data.

Postdoctoral Research Associate *(X-ray Scattering Signal Modeling)* Jan 2025 – Present

Northeastern University, Dept. of Bioengineering, Boston, MA

- Developed a correlation-aware feature reduction method to extract weak biological signatures from noisy X-ray scattering data, reducing 211 candidate features to 11 disease-discriminative markers.
- Derived Bayes-risk and approximation-error bounds that determine pruning limits for highly correlated data while preserving discriminative signal.
- Trained a compact, interpretable neural network with 174 parameters to estimate pathological protein aggregation, achieving 84.3% test F1-score on held-out data.

Graduate Research Assistant *(Molecular Mapping of Alzheimer's Pathology)* Oct 2020 – Dec 2024

Northeastern University, Dept. of Electrical & Computer Engineering, Boston, MA

- Designed an inverse-problem framework that generates Alzheimer's pathology maps from raw X-ray microdiffraction data, partnering with wet-lab teams at MGH for histology and biophysicists at Brookhaven National Laboratory for synchrotron measurements across 15 human brain samples.
- Formulated a pipeline that chains background subtraction, t-SNE, Gaussian Mixture Models, and a neural network to generate probabilistic spatial maps of pathological protein aggregates from noisy X-ray scattering measurements.
- Identified a consistent molecular signature of Alzheimer's cross-β fibrils: a $\sim$4.7 Å β-strand spacing with an approximately -1° left-handed twist that persisted across disease stages.
- Authored 5 peer-reviewed publications and delivered 5 conference presentations, including 2 invited talks recognized with student lecturer awards.

Visiting Graduate Researcher *(Cryo-ET Automation)* Sep 2022 – Dec 2022

Harvard Medical School, Dept. of Biological Chemistry & Molecular Pharmacology, Boston, MA

- Designed a Python-based GUI to automate cryo-electron tomography data acquisition, including microscope setup, motion correction, drift correction, and tilt-series alignment.
- Reduced setup time by 40% and increased throughput for structural biology workflows at Boston-area hospitals including Boston Children's Hospital, Dana-Farber, and MGH.

Ancillary Co-Principal Investigator *(Multi-Omics Biomarkers of Lung Disease)* Nov 2025 – Present
Brigham and Women's Hospital / Harvard Medical School, Boston, MA

- Lead transcriptomic analysis of COPDGene participants ($n \approx 3{,}000$) to identify molecular signatures associated with mechanically affected lung (MAL) using participant stratification, differential gene expression, and pathway enrichment analysis.
- Develop proteomics mediation models to identify plasma proteins that mediate the relationship between mechanically affected lung (MAL) and 5-year emphysema progression.

Graduate Research Assistant *(Wildlife Signal Detection and Sensing Device)* Jan 2018 – Jun 2019
Texas State University, Dept. of Electrical Engineering, San Marcos, TX

- Prototyped and deployed a solar-powered Raspberry Pi acoustic monitor with onboard inference and cellular alerting for endangered Houston toad detection, scaling the system to 50 field units for conservation use in Bastrop County, Texas.
- Implemented power spectral density analysis, tuned bandpass filtering, MFCC extraction, and an LSTM classifier to distinguish Houston toad mating calls from other animal sounds and field noise, achieving 98% true-positive and 92.6% test accuracy on field audio.

TECHNICAL SKILLS

Statistical Learning: Bayesian inference, stochastic modeling, spectral analysis, signal denoising

ML Methods: Graph neural networks, contrastive learning, semi-supervised learning, transformer models

ML Workflow Tools: Claude Code, W&B, Hugging Face

Frameworks & Libraries: PyTorch, scikit-learn, OpenCV, NumPy, SciPy

Compute & Version Control: SLURM/HPC, GitHub

Programming: Python, R, MATLAB, Bash, C/C++

EDUCATION

Ph.D. in Electrical Engineering Dec 2024
Northeastern University, Boston, MA
Advisor: Prof. Lee Makowski

M.Sc. in Electrical Engineering Jun 2019
Texas State University, San Marcos, TX
Advisors: Prof. Damian Valles and Prof. Michael Forstner

B.Sc. in Electrical & Electronic Engineering Jul 2011
Rajshahi University of Engineering & Technology, Bangladesh

SELECTED PUBLICATIONS

- **A. A. Bashit**, P. Nepal, and L. Makowski, "A Multicollinearity-Aware Signal-Processing Framework for Cross-β Identification via X-ray Scattering of Alzheimer's Tissue," *under review*, Nov. 2025. [arXiv]